



UNIVERSITÀ DEGLI STUDI DI MILANO
FACOLTÀ DI SCIENZE E TECNOLOGIE

Reinforcement-Learning Framework for Robust and Universal Analog Quantum Control on Superconducting Qubits

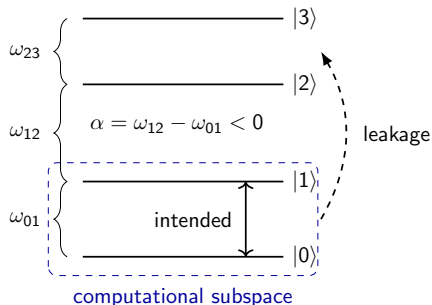
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Scope of the thesis

- For scalable and reliable quantum devices, high-fidelity, fast, and robust quantum gates are needed.
- Superconducting qubits are a leading platform, but makes control design challenging due to leakage and noise sensitivity.



Physical issue

Superconducting qubits are **weakly anharmonic**, so noise or fast control can easily excite unwanted higher levels. This phenomenon is called **leakage**.

Analog quantum control

Implementation of a quantum operation by injecting control pulses at the Hamiltonian level.

Why this problem matters

Main difficulty

In superconducting qubits, gate design is not only a matter of logical accuracy. We must balance several competing goals:

- **high gate fidelity**
- **short runtime**
- **low leakage** outside the computational subspace
- **robustness** against control noise

Why Deep Reinforcement Learning (RL)?

Sutton, R. S. and Barto, A. G., *Reinforcement Learning: An Introduction*, MIT press (2018). · Koch, C. P., *Quantum Optimal Control in Quantum Technologies*, EPJ QT (2022).

Thesis goals and main contribution

Scientific goal

Assess whether the framework introduced in the 2019 Nature paper *Universal Quantum Control through Deep Reinforcement Learning* by Niu *et al.* is a credible strategy for **robust analog quantum control** on simulated gmon qubits.

What I specifically did

Significant effort was spent on implementation: no public code was available, the whole framework had to be rebuilt from scratch. This included:

- Quantum Environment simulator of two-qubit gmon architecture
- Leakage estimator
- Full Deep Reinforcement Learning pipeline (neural networks, training, evaluation), coupled with Trust Region Policy Optimization (PyTorch)
- Optimization to make the framework computationally feasible
- Experiment design and independent validation

Niu, M. Y. et al., *Universal Quantum Control through Deep Reinforcement Learning*, EPJ Quantum Technology (2019).

Physical model: gmon Hamiltonian and control channels

The effective Hamiltonian is

$$\hat{H}(t) = \underbrace{\frac{\eta}{2} \sum_{j=1}^2 \hat{n}_j(\hat{n}_j - 1)}_{\text{anharmonicity}} + \underbrace{g(t)(\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2)}_{\text{tunable coupling}} + \underbrace{\sum_{j=1}^2 \delta_j(t) \hat{n}_j}_{\text{local detuning}} + \underbrace{\sum_{j=1}^2 i f_j(t) (\hat{a}_j e^{i\phi_j(t)} - \hat{a}_j^\dagger e^{-i\phi_j(t)})}_{\text{microwave drive}}$$

Control vector: Agent's output

The control vector is:

$$a_t = (g(t), \delta_1(t), \delta_2(t), f_1(t), f_2(t), \phi_1(t), \phi_2(t))$$

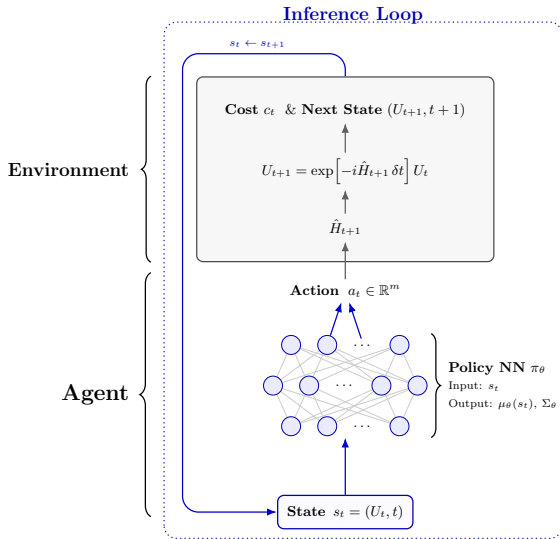
And this is going to be the output of the RL agent at each time step.

Solution to the control problem

A solution to the control problem is a sequence of control vectors:

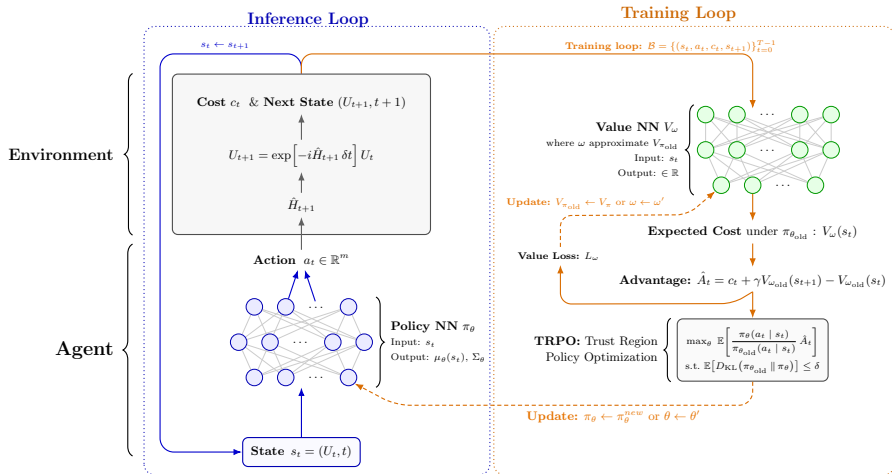
$$\{a_t\}_{t=0}^T$$

RL formulation: Inference Loop



Sutton, R. S. and Barto, A. G., *Reinforcement Learning: An Introduction*, MIT press (2018). · Niu, M. Y. et al., *Universal Quantum Control through Deep Reinforcement Learning*, EPJ QT (2019).

RL formulation: complete



Sutton, R. S. et al., *Reinforcement Learning: An Introduction* (2018). · Niu, M. Y. et al., *Universal Quantum Control...* (2019). · Schulman, J. et al., *Trust Region Policy Optimization* (2015).

Why this is a multi-objective control problem

Cost Function and Multi-objective nature

The cost function is the core metric that the agent is trained to minimize. Therefore, the multi-objective nature of the control problem must be explicitly encoded in the cost function.

Universal Function Optimizaion (UFO) cost function

$$C = \underbrace{\chi[1 - F(U(T))]}_{\text{fidelity penalty}} + \underbrace{\beta L_{\text{tot}}(T)}_{\text{leakage penalty}} + \underbrace{\mu P_{\text{bdry}}}_{\text{boundary penalty}} + \underbrace{\kappa T}_{\text{time penalty}}$$

- These goals are not automatically compatible: reducing one penalty can worsen another.

Experiments performed

Controller variants implemented

- **Classical Optimization:** Optimizing the $\{a_t\}_{t=0}^T$ to minimize the UFO cost function via Gradient Descent (**ADAM**).
- **Reinforcement-Learning Agent** trained in multiple configurations.

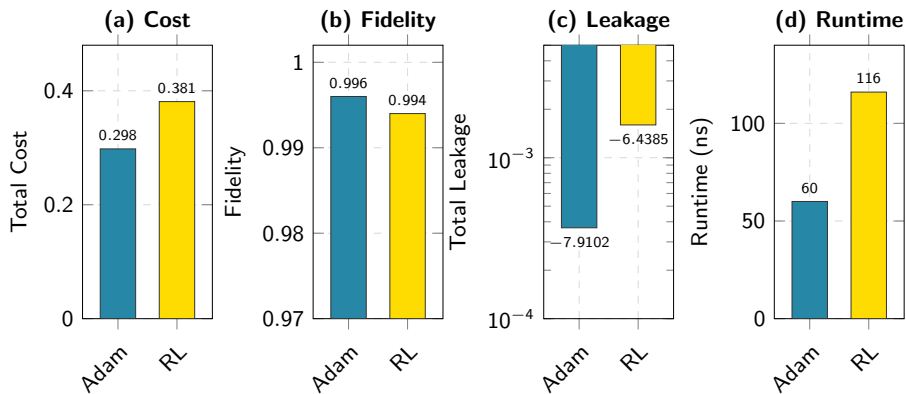
Main experiments performed

- 1 **Extensive parameter study:**
 - neural network sizes and layouts
 - UFO cost function weights
- 2 **Single-target benchmark:** Adam vs RL Agent
- 3 **Robustness under noise:** fidelity metrics vs noise strength
- 4 **Consistency of performance:** RL Agent vs Adam optimization
- 5 **Transfer Learning Evaluation:** across nearby targets

Single-target optimization: Nominal RL agent vs Adam

Interpretation

In the nominal single-target regime, the direct Adam baseline remains stronger.

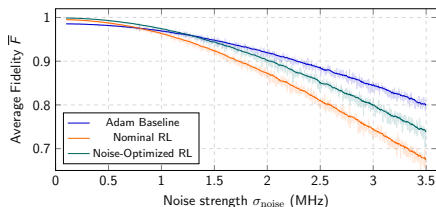


Noise-optimized Agent: Robustness

Under control noise, the **noise-optimized RL agent** (trained with 1MHz noise) always beats **nominal RL**: the gain comes from **training in a stochastic environment**, not from RL alone. Its lead over the noise-aware **Adam** baseline holds only near the training noise.

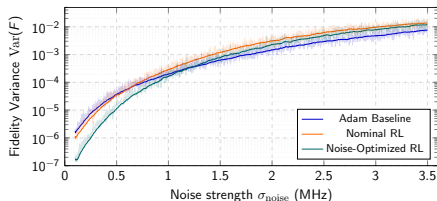
Average fidelity

- above nominal RL at every σ_{noise}
- above Adam only up to $\sigma_{\text{noise}} \approx 1.3 \text{ MHz}$; Adam leads beyond



Fidelity variance

- lower than nominal RL at every σ_{noise}
- lowest overall only up to $\approx 1.2 \text{ MHz}$; Adam is most stable beyond



Conclusions

What is not fully shown in this presentation

Although I only presented the main results, the thesis also includes several additional analyses:

- extended robustness comparisons under stochastic control noise
- runtime analysis across the gate family for both the baseline and the RL controller
- sensitivity studies on the UFO cost-function weights
- neural-network architecture sweeps
- transfer-learning behavior under curriculum training
- exploration of training and simulation parameter choices

What this thesis contributed

- full reconstruction of the framework from scratch
- extended existing literature with a comprehensive comparison between RL and a strong classical technique.

Scientific conclusions

- Under stochastic perturbations, noise-optimized RL beats nominal RL everywhere, but beats the noise-aware Adam baseline only near its training noise.

Additional materials and results follow

Additional work beyond what's shown here

What I had to build and learn

To carry out this thesis, I had to reconstruct the full framework from scratch and acquire working knowledge of:

- Multilevel quantum simulation, PyTorch, autograd
- classical optimization and both classical and deep reinforcement learning
- policy-gradient methods and neural-network training
- superconducting-qubit physics and quantum control

Universal Function Optimizaion (UFO) cost function

$$C = \underbrace{\chi[1 - F(U(T))]}_{\text{fidelity penalty}} + \underbrace{\beta L_{\text{tot}}(T)}_{\text{leakage penalty}} + \underbrace{\mu P_{\text{bdry}}}_{\text{boundary penalty}} + \underbrace{\kappa T}_{\text{time penalty}}$$

Fidelity

Being $U(T)$ the unitary evolution generated by a control protocol at final time T , and U_{target} the desired logical gate.

$$F(U(T)) = \frac{1}{d_{\text{comp}}^2} \left| \text{Tr} \left(U^\dagger(T) U_{\text{target}} \right) \right|^2, \quad \text{with } d_{\text{comp}} = 4$$

Boundary Penalty

$$P_{\text{bdry}} = \sum_{t \in \{0, T\}} [g(t)^2 + f_1(t)^2 + f_2(t)^2]$$

where $g(t)$, $f_1(t)$, $f_2(t)$ are the amplitude controllers at a given time t . This term forces (by penalizing) the control amplitude to be zero at the pulse boundaries, which can cause unwanted excitations and leakage after the termination of the episode.

Leakage term: physics-informed proxy

Considering the separated Hamiltonian in block form, $\hat{H}_{\text{od}}(t)$ is the off-diagonal block that couples the computational subspace to the leakage subspace, while $\Delta(t)$ is the instantaneous energy gap between these subspaces.

$$L_{\text{tot}}(T) \lesssim \frac{\|\hat{H}_{\text{od}}(0)\|}{\Delta(0)} + \frac{\|\hat{H}_{\text{od}}(T)\|}{\Delta(T)} + \int_0^T \frac{1}{\Delta(t)^2} \left\| \frac{d^2 \hat{H}_{\text{od}}(t)}{dt^2} \right\| dt$$

The first two terms penalize residual couplings at the pulse boundaries; the integral term penalizes rapid temporal variations that can induce non-adiabatic transitions, effectively working as a smoothing filter.

Time penalty

Since the experiment values do not coincide with the natural time units of the system, a normalization factor is introduced.

$$T = \frac{t}{T_{\text{norm}}}, \quad \text{with } T_{\text{norm}} = 60 \text{ ns}$$

Chosen target gate family

Target gate family used throughout the numerical study.

$$N(\alpha, \alpha, \gamma) = \exp \left[i \left(\alpha \sigma_1^x \sigma_2^x + \alpha \sigma_1^y \sigma_2^y + \gamma \sigma_1^z \sigma_2^z \right) \right].$$

Evaluation metrics & target gate family

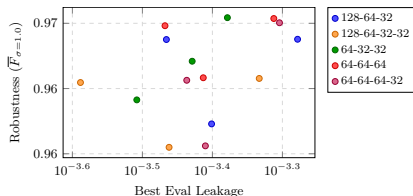
Metric	Definition	Physical meaning
Gate fidelity	$F_{\text{gate}} = \frac{1}{16} \left \text{Tr} \left(U_{\text{proj}}^\dagger(T) U_{\text{target}} \right) \right ^2$	Logical accuracy of the implemented two-qubit gate.
Leakage proxy	$L_{\text{tot}}(T) \lesssim \frac{\ \hat{H}_{\text{od}}(0)\ }{\Delta(0)} + \frac{\ \hat{H}_{\text{od}}(T)\ }{\Delta(T)} + \int_0^T \frac{1}{\Delta(t)^2} \left\ \frac{d^2 \hat{H}_{\text{od}}(t)}{dt^2} \right\ dt$	Estimated coherent population loss outside the computational subspace.
Average fidelity under noise	$\bar{F}(\sigma) = \mathbb{E}[F]$	Mean gate performance over many noisy Monte Carlo realizations.
Fidelity variance	$\text{Var}(F) = \mathbb{E}[(F - \bar{F})^2]$	Shot-to-shot stability under stochastic perturbations.

Extended parameter study

In RL, parameter choice strongly affects stability, convergence, robustness, and final control quality. I performed an extended study on:

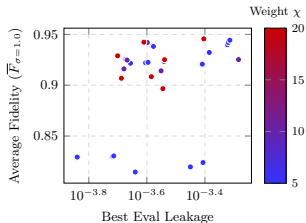
Policy/value architectures

- sweep over 5 network topologies
- final choice: (64, 32, 32)



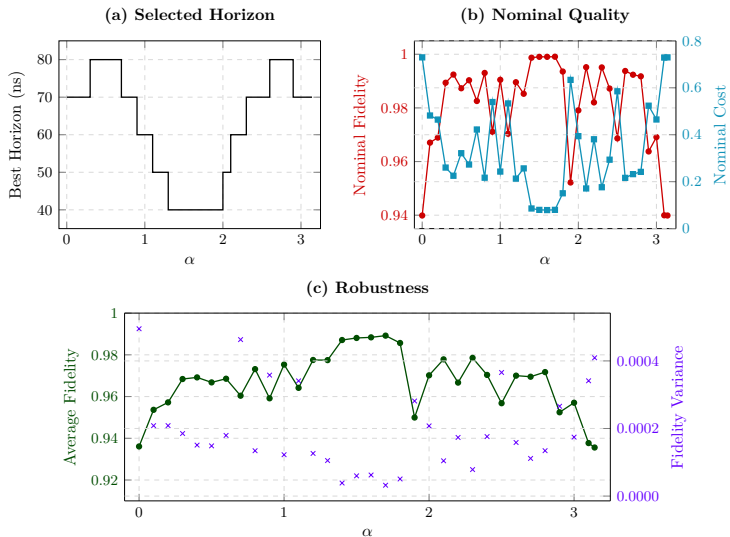
UFO weights

- sensitivity study over 36 configurations
- final choice: (10, 10, 0.2, 0.1)



Adam vs TRPO and Family-wide evaluation

Family-wide runtime sweep was also evaluated to mostly assess performance consistency across the target gate family of the Adam baseline.



Runtime across the gate family $N(\alpha, \alpha, \pi/2)$

Target parametrization: $U_{\text{target}}(\alpha) = N(\alpha, \alpha, \pi/2)$. The runtime landscape is **highly structured**, not uniform.

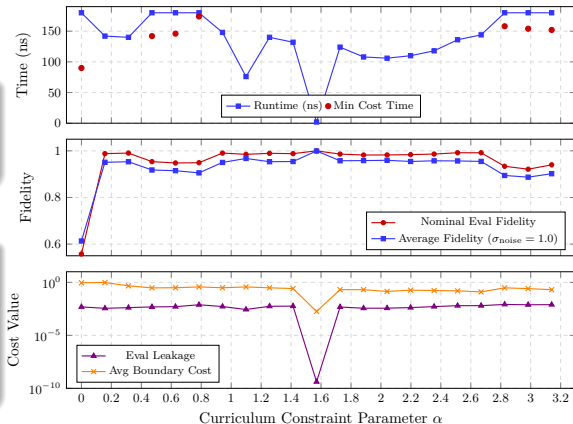
- A broad central region admits significantly faster analog solutions.
- The analog controller stays below the gate-synthesis reference over a large part of the family.

Reference benchmark

$T_{\text{syn}} \approx 215 \text{ ns}$ from optimal discrete gate synthesis.

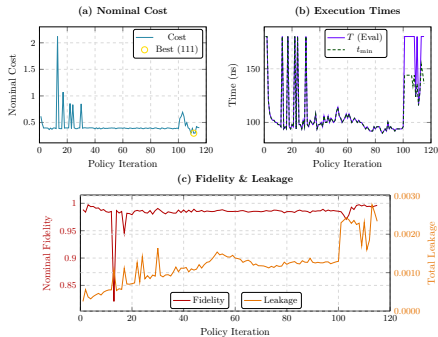
Physical interpretation

The strongest speedups appear where the target gate is better aligned with the native hardware interaction.

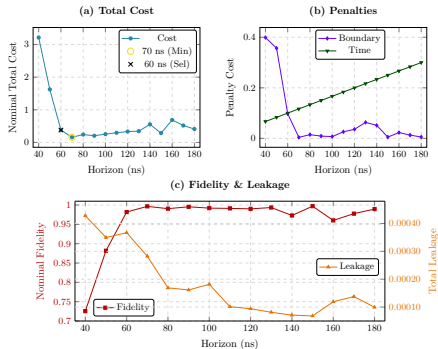


Single-target: Training Dynamics and Horizon Sweep

RL Agent training dynamics



Single-target horizon sweep



Current propagator and state definition

The current propagated unitary is

$$U_t = \mathcal{T} \exp \left(-i \int_0^t H(\tau) d\tau \right),$$

so it represents the **time-evolution operator accumulated up to time t** .

In the implementation, the RL state is built from the current propagator as

$$s_t = \left(\text{Re}(\text{vec}(U_t)), \text{Im}(\text{vec}(U_t)), t/T_{\text{norm}} \right).$$

With a three-level truncation for each mode:

$$d_{\text{phys}} = 3^2 = 9 \quad \implies \quad U_t \in \mathbb{C}^{9 \times 9},$$

so the observation dimension is

$$d_{\text{obs}} = 2 \cdot 9^2 + 1 = 163.$$

Policy network

Input: current state s_t

Output: parameters of a Gaussian policy

$$\pi_{\theta}(a_t | s_t) = \mathcal{N}(\mu_{\theta}(s_t), \Sigma_{\theta}),$$

with

$$\mu_{\theta}(s_t) \in \mathbb{R}^7, \quad \Sigma_{\theta} = \text{diag}(\sigma_1^2, \dots, \sigma_7^2).$$

After sampling and squashing:

$$a_t = \tanh(\tilde{a}_t) \in [-1, 1]^7, \text{ where } \tilde{a}_t = \mu_{\theta}(s_t) + \Sigma_{\theta}^{1/2} \epsilon.$$

then rescaled to the physical controls

$$u_t = (g_t, \delta_{1,t}, \delta_{2,t}, f_{1,t}, f_{2,t}, \phi_{1,t}, \phi_{2,t}).$$

Value network

Value network

Input: current state s_t

Output: scalar state value

$$V_{\omega}(s_t) \in \mathbb{R}.$$

This estimates the expected discounted future reward:

$$V_{\omega}(s_t) \approx \mathbb{E} \left[\sum_{k=t}^T \gamma^{k-t} r_k \mid s_t \right].$$

Interpretation:

- the **policy** proposes the next control action,
- the **value network** estimates how promising the current state is.

References I



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